Experiment 9

AIM: Use pytest to test a Python application.

→Create Project PythonDemo and add file pythondemo.py

```
C:\Users\ADMIN\OneDrive\Desktop\SEM 6\OSTLAB>pip install pytest
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pytest in
c:\users\admin\appdata\local\packages\pythonsoftwarefoundation.python.3.13_qbz5n2kfr
a8p0\localcache\local-packages\python313\site-packages (9.0.2)
Requirement already satisfied: colorama>=0.4 in
c:\users\admin\appdata\local\packages\pythonsoftwarefoundation.python.3.13_qbz5n2kfr
a8p0\localcache\local-packages\python313\site-packages (from pytest) (0.4.6)
Requirement already satisfied: iniconfig>=1.0.1 in
c:\users\admin\appdata\local\packages\pythonsoftwarefoundation.python.3.13_qbz5n2kfr
a8p0\localcache\local-packages\python313\site-packages (from pytest) (2.3.0)
Requirement already satisfied: packaging>=22 in
c:\users\admin\appdata\local\packages\pythonsoftwarefoundation.python.3.13_qbz5n2kfr
a8p0\localcache\local-packages\python313\site-packages (from pytest) (26.0)
Requirement already satisfied: pluggy<2,>=1.5 in
c:\users\admin\appdata\local\packages\pythonsoftwarefoundation.python.3.13_qbz5n2kfr
a8p0\localcache\local-packages\python313\site-packages (from pytest) (1.6.0)
Requirement already satisfied: pygments>=2.7.2 in
c:\users\admin\appdata\local\packages\pythonsoftwarefoundation.python.3.13_qbz5n2kfr
a8p0\localcache\local-packages\python313\site-packages (from pytest) (2.20.0)

[notice] A new release of pip is available: 25.3 -> 26.0.1
[notice] To update, run:
C:\Users\ADMIN\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3
.13_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip

C:\Users\ADMIN\OneDrive\Desktop\SEM 6\OSTLAB>pytest --version
pytest 9.0.2

C:\Users\ADMIN\OneDrive\Desktop\SEM 6\OSTLAB>mkdir PythonDemo

C:\Users\ADMIN\OneDrive\Desktop\SEM 6\OSTLAB>cd PythonDemo

C:\Users\ADMIN\OneDrive\Desktop\SEM 6\OSTLAB\PythonDemo>notepad pythondemo.py
```

pythondemo.py

```
def add(a, b):
    return a + b

def subtract(a, b):
    return a - b

def multiply(a, b):
    return a * b

def divide(a, b):
    if b == 0:
        raise ValueError("Cannot divide by zero")
```

```
    return a / b

def is_even(num):
    return num % 2 == 0

C:\Users\ADMIN\OneDrive\Desktop\SEM 6\OSTLAB\PythonDemo>notepad test_pythondemo.py
test_pythondemo.py
import pytest
import pythondemo

# Passing test
def test_add():
    assert pythondemo.add(2, 3) == 5

# Failing test
def test_multiply_fail():
    assert pythondemo.multiply(4, 3) == 11

# Exception test
def test_divide_by_zero():
    with pytest.raises(ValueError):
        pythondemo.divide(10, 0)

# Logical test
def test_is_even():
    assert pythondemo.is_even(4) is True
    assert pythondemo.is_even(5) is False

# Parameterized test
@pytest.mark.parametrize("a,b,result", [
    (1,2,3),
    (2,2,4),
    (5,5,11)  # failing case
])
def test_add_param(a,b,result):
    assert pythondemo.add(a,b) == result

C:\Users\ADMIN\OneDrive\Desktop\SEM 6\OSTLAB\PythonDemo>pytest -v
================================ test session starts ================================
platform    win32    --    Python    3.13.12,    pytest-9.0.2,    pluggy-1.6.0    --
C:\Users\ADMIN\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3
.13_qbz5n2kfra8p0\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\ADMIN\OneDrive\Desktop\SEM 6\OSTLAB\PythonDemo
collected 7 items

test_pythondemo.py::test_add PASSED                                          [ 10%]
test_pythondemo.py::test_multiply_fail FAILED                                [ 20%]
test_pythondemo.py::test_divide_by_zero PASSED                               [ 30%]
test_pythondemo.py::test_is_even PASSED                                      [ 40%]
test_pythondemo.py::test_add_param[1-2-3] PASSED                             [ 50%]
test_pythondemo.py::test_add_param[2-2-4] PASSED                             [ 60%]
test_pythondemo.py::test_add_param[5-5-11] FAILED                            [ 70%]
=================================FAILURES=================================
test_multiply_fail
```

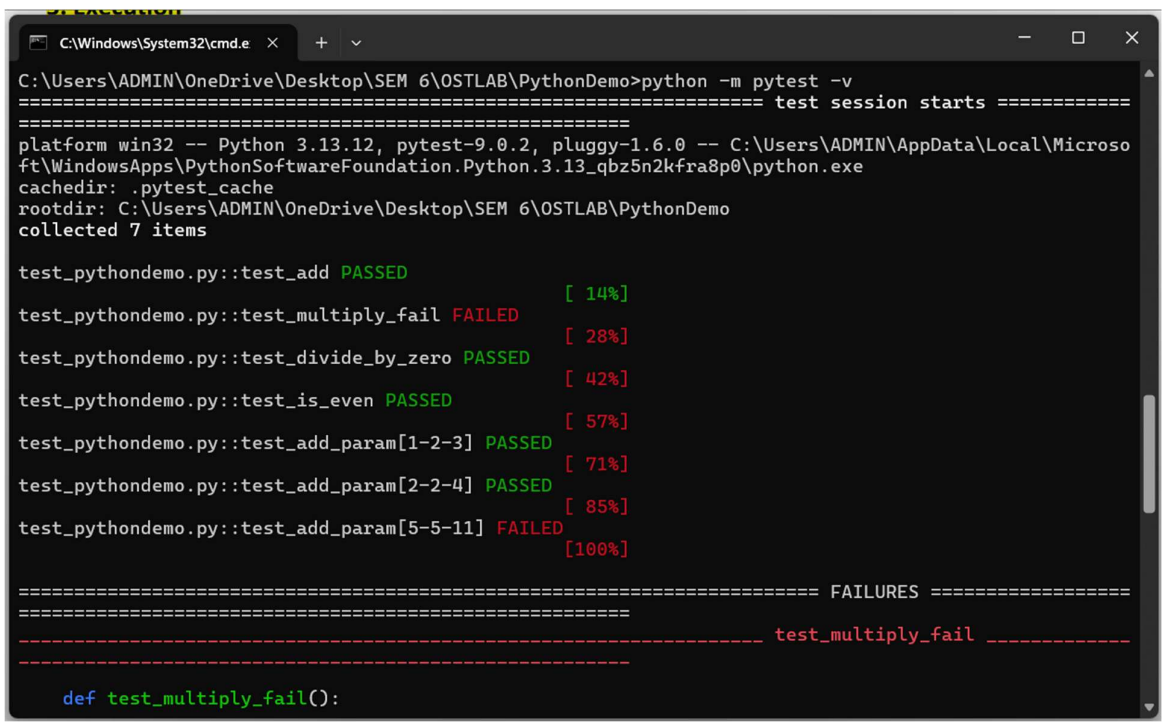
```
    def test_multiply_fail():
>       assert pythondemo.multiply(4, 3) == 11
E       assert 12 == 11
E        +  where 12 = <function multiply at 0x000001F45C7E7060>(4, 3)
E        +    where <function multiply at 0x000001F45C7E7060> = pythondemo.multiply

test_pythondemo.py:10: AssertionError
_____________________________ test_add_param[5-5-11] _____________________________

a = 5, b = 5, result = 11

    @pytest.mark.parametrize("a,b,result", [
        (1,2,3),
        (2,2,4),
        (5,5,11)  # failing case
    ])
    def test_add_param(a,b,result):
>       assert pythondemo.add(a,b) == result
E       assert 10 == 11
E        +  where 10 = <function add at 0x000001F45C7E6F20>(5, 5)
E        +    where <function add at 0x000001F45C7E6F20> = pythondemo.add

test_pythondemo.py:29: AssertionError
============================= warnings summary =============================
```



```
_____________________________________________________________ test_multiply_fail _____________
____________________________________________________

    def test_multiply_fail():
>       assert pythondemo.multiply(4, 3) == 11
E       assert 12 == 11
E        +  where 12 = <function multiply at 0x000001F45C7E7060>(4, 3)
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test_pythondemo.py:10: AssertionError
_____________________________________________________________ test_add_param[5-5-11] ___________
____________________________________________________

a = 5, b = 5, result = 11

    @pytest.mark.parametrize("a,b,result", [
        (1,2,3),
        (2,2,4),
        (5,5,11)  # failing case
    ])
    def test_add_param(a,b,result):
>       assert pythondemo.add(a,b) == result
E       assert 10 == 11
E        +  where 10 = <function add at 0x000001F45C7E6F20>(5, 5)
E        +    where <function add at 0x000001F45C7E6F20> = pythondemo.add

test_pythondemo.py:29: AssertionError
================================================================ warnings summary ==============
____________________________________________________
```

Task 1: Add factorial function

ADD in pythondemo.py:

```
def factorial(n):
    if n < 0:
        raise ValueError("Negative not allowed")
    if n == 0:
        return 1
    return n * factorial(n-1)
```

Add test for factorial:

```
def test_factorial():
    assert pythondemo.factorial(5) == 120
```

Task 2: Edge cases

```
def test_negative():
    with pytest.raises(ValueError):
        pythondemo.factorial(-1)

def test_large():
    assert pythondemo.factorial(6) == 720
```

CMD for Testing:

```
C:\Users\ADMIN\OneDrive\Desktop\SEM 6\OSTLAB\PythonDemo>python -m pytest -v
========================== test session starts ====================================
platform    win32    --    Python    3.13.12,    pytest-9.0.2,    pluggy-1.6.0    --
C:\Users\ADMIN\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3
.13_qbz5n2kfra8p0\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\ADMIN\OneDrive\Desktop\SEM 6\OSTLAB\PythonDemo
collected 10 items
```

```
test_pythondemo.py::test_add PASSED                                    [ 10%]
test_pythondemo.py::test_multiply_fail FAILED                          [ 20%]
test_pythondemo.py::test_divide_by_zero PASSED                         [ 30%]
test_pythondemo.py::test_is_even PASSED                                [ 40%]
test_pythondemo.py::test_add_param[1-2-3] PASSED                       [ 50%]
test_pythondemo.py::test_add_param[2-2-4] PASSED                       [ 60%]
test_pythondemo.py::test_add_param[5-5-11] FAILED                      [ 70%]
test_pythondemo.py::test_factorial PASSED                              [ 80%]
test_pythondemo.py::test_negative PASSED                               [ 90%]
test_pythondemo.py::test_large PASSED                                  [100%]
```

```
C:\Users\ADMIN\OneDrive\Desktop\SEM 6\OSTLAB\PythonDemo>python -m pytest -v
========================================= test session starts =========================================
platform win32 -- Python 3.13.12, pytest-9.0.2, pluggy-1.6.0 -- C:\Users\ADMIN\AppData\Local\Microso
ft\WindowsApps\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\ADMIN\OneDrive\Desktop\SEM 6\OSTLAB\PythonDemo
collected 10 items

test_pythondemo.py::test_add PASSED                                                          [ 10%]
test_pythondemo.py::test_multiply_fail FAILED                                                [ 20%]
test_pythondemo.py::test_divide_by_zero PASSED                                               [ 30%]
test_pythondemo.py::test_is_even PASSED                                                      [ 40%]
test_pythondemo.py::test_add_param[1-2-3] PASSED                                             [ 50%]
test_pythondemo.py::test_add_param[2-2-4] PASSED                                             [ 60%]
test_pythondemo.py::test_add_param[5-5-11] FAILED                                            [ 70%]
test_pythondemo.py::test_factorial PASSED                                                    [ 80%]
test_pythondemo.py::test_negative PASSED                                                     [ 90%]
test_pythondemo.py::test_large PASSED                                                        [100%]
```